

The Direct Medial and Lateral Inferior Horizontal Portals of the Knee: A Modified Arthroscopic Approach for Anterior Horn Horizontal Tears

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Abstract: Mild horizontal tears in the meniscal avascular zone are common intraoperative findings during complex meniscal or discoid meniscus surgeries. However, resecting the inferior surface of such tears in the anterior one-half to one-third of the meniscal body is difficult and time-consuming because of arthroscopic visualization and instrument limitations. Herein, we describe a direct medial and lateral inferior portal inferior to the anterior one-third of the meniscal body, which enables efficient debridement of deep horizontal tears with shavers or rongeurs and enhances arthroscopic observation, thus improving surgical efficiency and reducing procedural difficulty.

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As a critical structure of the knee joint, the meniscus plays a pivotal role in maintaining the joint's function, stability, and nutritional supply. However, its unique structural and functional attributes make it highly susceptible to injuries during physical activities.¹ When meniscal injuries are accompanied by significant knee pain and impaired joint function, surgical intervention is often required. Among the various types of meniscal injuries, horizontal tears represent a relatively specific entity. Isolated horizontal tears are often a consequence of meniscal degeneration and can be managed conservatively if symptoms are mild.² However, during surgical treatment for complex meniscal tears or symptomatic discoid menisci (e.g., bucket-handle tears caused by discoid menisci), mild horizontal tears located in the avascular zone (white zone) are also frequently identified intraoperatively.^{3,4} In such cases, because these tears do not disrupt the annular collagen fiber structure of the meniscus, direct resection of the inferior surface is typically performed to

prevent potential residual symptoms after surgery.⁵ However, although this management strategy is extremely simple and efficient in itself, when the horizontal tear is located precisely in the anterior one-half to one-third region of the meniscal body, resection of its inferior surface becomes extremely difficult and time-consuming, constrained by the limitations of arthroscopic visualization and the operational reach of instruments (primarily rongeurs). This not only greatly compromises surgical efficiency but also may preclude the complete removal of the unstable inferior meniscal surface (Figure 1).

Therefore, we describe a modified arthroscopic approach located inferior to the anterior one-third of the meniscal body, specifically designed to address this issue. Through this approach, shavers or rongeurs can be rapidly used to debride deep horizontal meniscal tears, and more detailed arthroscopic observation can be performed, thereby significantly improving surgical efficiency and reducing surgical difficulty.

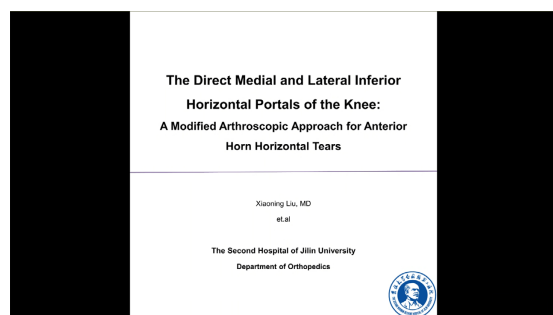
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VIDEO 1 Synopsis of the modified submeniscal portal technique for the debridement of deep horizontal meniscal tears. The video provides a step-by-step demonstration of the surgical procedure. Anatomical illustrations are presented to highlight the core principle of portal placement: using a “safe incision zone” at the joint line that navigates anterior to the collateral ligaments (avoiding the lateral collateral and anterolateral ligaments laterally and the medial collateral ligament medially) to safely access the submeniscal working zone. The surgical footage then shows the practical application of this technique on a medial meniscus. The key surgical steps include (1) percutaneous localization using a puncture needle, precisely entering the space between the inferior surface of the meniscus and the tibial articular surface; (2) making a superficial skin incision followed by meticulous blunt dissection down to the joint capsule to protect surrounding neurovascular and ligamentous structures; and (3) establishing the portal under direct arthroscopic monitoring to introduce a shaver along this specific trajectory, allowing for the thorough and safe debridement of the deep horizontal tear.

Video content can be viewed at <https://doi.org/10.1002/atn.70283>.

SURGICAL TECHNIQUE

A detailed demonstration of the surgical procedure is presented in Video 1. Spinal anesthesia was administered for the surgery. Initially, routine anterolateral and antero-medial arthroscopic portals are used for exploration and debridement. When the horizontal tear is located in the posterior half of the meniscal body or more posteriorly, conventional arthroscopic portals allow for easy visualization and manipulation. In contrast, when the tear is situated in the anterior half of the meniscal body, our modified submeniscal approach (direct medial and lateral inferior portals) is highly recommended.

Before establishing the portals, the disparate anatomical structures and compartment laxity of each side must be carefully considered. The lateral compartment inherently possesses less laxity than the medial side, necessitating precise portal placement to protect lateral

stabilizing structures. As the lateral collateral ligament is located posterolaterally and the anterolateral ligament is located anterolaterally, the optimal entry point for the lateral portal is pinpointed at the midlateral edge of the tibial plateau (Figure 2A,B). This specific safe zone provides an excellent trajectory and sufficient working space to address the entire anterior half of the meniscus while safely navigating between these crucial structures.

To establish the portal, arthroscopic observation is first performed from the contralateral side of the injured meniscus. After confirming the location of the horizontal tear, a long puncture needle is percutaneously inserted into the joint cavity, entering the space directly inferior to the delaminated meniscus and superior to the tibial plateau (Figure 3A,B). The needle must be kept strictly parallel to the tibial plateau under direct arthroscopic visualization to avoid iatrogenic damage to the meniscal parenchyma or articular cartilage. If the needle position is unsatisfactory, it must be readjusted. For beginners, the arthroscopic light spot on the skin surface can serve as a reference for the entry area, or 2 needles placed 5 mm apart can be used to confirm the correct plane.

Crucially, to prevent iatrogenic injury to the ligamentous structures (the medial collateral ligament medially and the lateral collateral ligament and anterolateral ligament laterally), a meticulous surgical technique is required. Using the properly positioned puncture needle as a landmark, a sharp scalpel is used to make only a superficial, vertical skin incision (approximately 5 mm). Subsequently, blunt dissection (e.g., using a mosquito hemostat or a blunt trocar) is strictly employed to penetrate the joint capsule and enter the submeniscal space. This blunt separation technique effectively spreads the capsular and ligamentous fibers rather than transecting them, thereby ensuring the structural integrity of the knee joint.

After successful portal establishment, shavers or rongeurs can be introduced through this inferior portal to debride the deep horizontal meniscal tears (Figure 3C,D). During this procedure, the routine anterior portals serve as viewing portals. Furthermore, on completion of debridement, this newly established portal can also be used as a viewing portal. By inserting the arthroscope through this inferior portal, the meniscal parenchyma is situated superiorly and the tibial plateau inferiorly. Adjusting the viewing angle allows for excellent evaluation of the anterior horn, posterior horn, and meniscal root from an inferior perspective. Pearls and pitfalls of this modified technique are presented in Table 1.

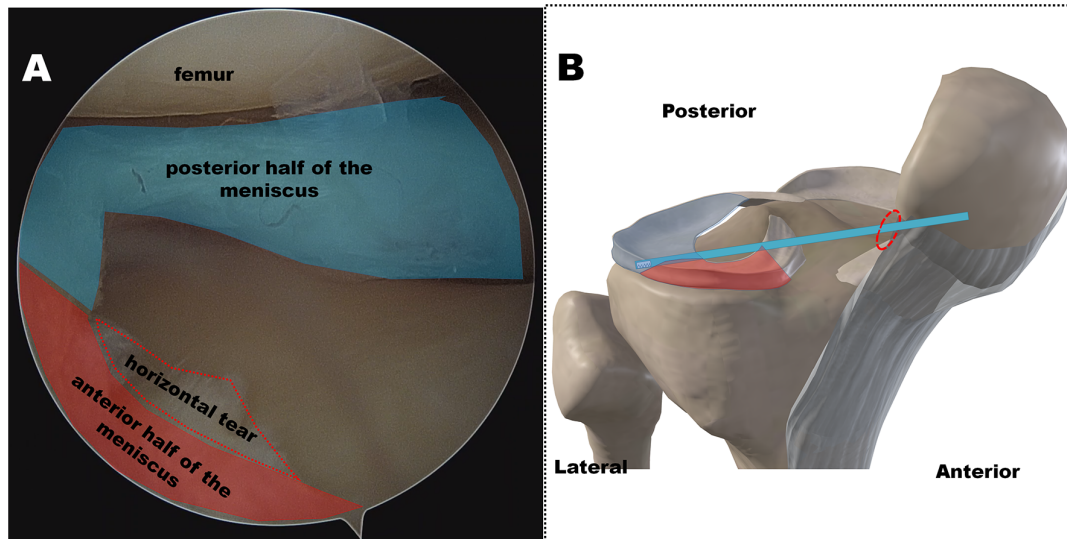


FIGURE 1 As shown in (A), when managing deep horizontal tears (marked by the red dashed line) in the anterior body and anterior horn of the meniscus (the red area), instruments such as arthroscopic shavers inserted through conventional anteromedial and anterolateral portals are difficult to access this region effectively because of limitations in their size and operating angle (B).

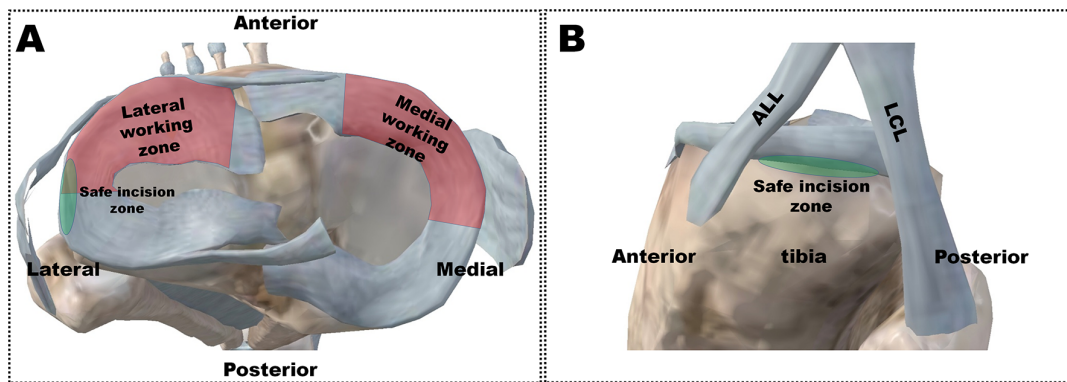


FIGURE 2 Anatomical landmarks and safe zones for establishing the direct inferior portals. (A) Superior view of the tibial plateau showing the proposed medial and lateral working zones (red areas), which strictly target the anterior half of the menisci. The safe incision zone for the lateral portal (green area) is pinpointed at the midlateral edge of the tibial plateau. (B) Lateral view of the knee joint illustrating the precise location of the safe incision zone (green area) relative to the lateral stabilizing structures. This optimal entry point safely navigates between the tibial insertion of the ALL anteriorly and the LCL posteriorly, ensuring sufficient working space while preventing iatrogenic injury during portal placement and subsequent blunt dissection. (ALL, anterolateral ligament; LCL, lateral collateral ligament.)

Ethics Statement

Written informed consent has been obtained from the patient to publish this article, and the patient has consented for all images and clinical data and other data included in the manuscript to be published. The patient provided informed consent to participate in the study. The patient provided informed consent for the publication of his anonymized case details and images.

DISCUSSION

For small horizontal tears in the avascular zone (white zone) of the meniscus identified intraoperatively, direct resection is often preferred given its minimal impact on the annular collagen fiber.⁶ However, when such injuries are located in the anterior half of the meniscal body or the anterior horn, conventional portals often create a dilemma of limited visualization and restricted instrument reach. Addressing this typically requires specialized surgical

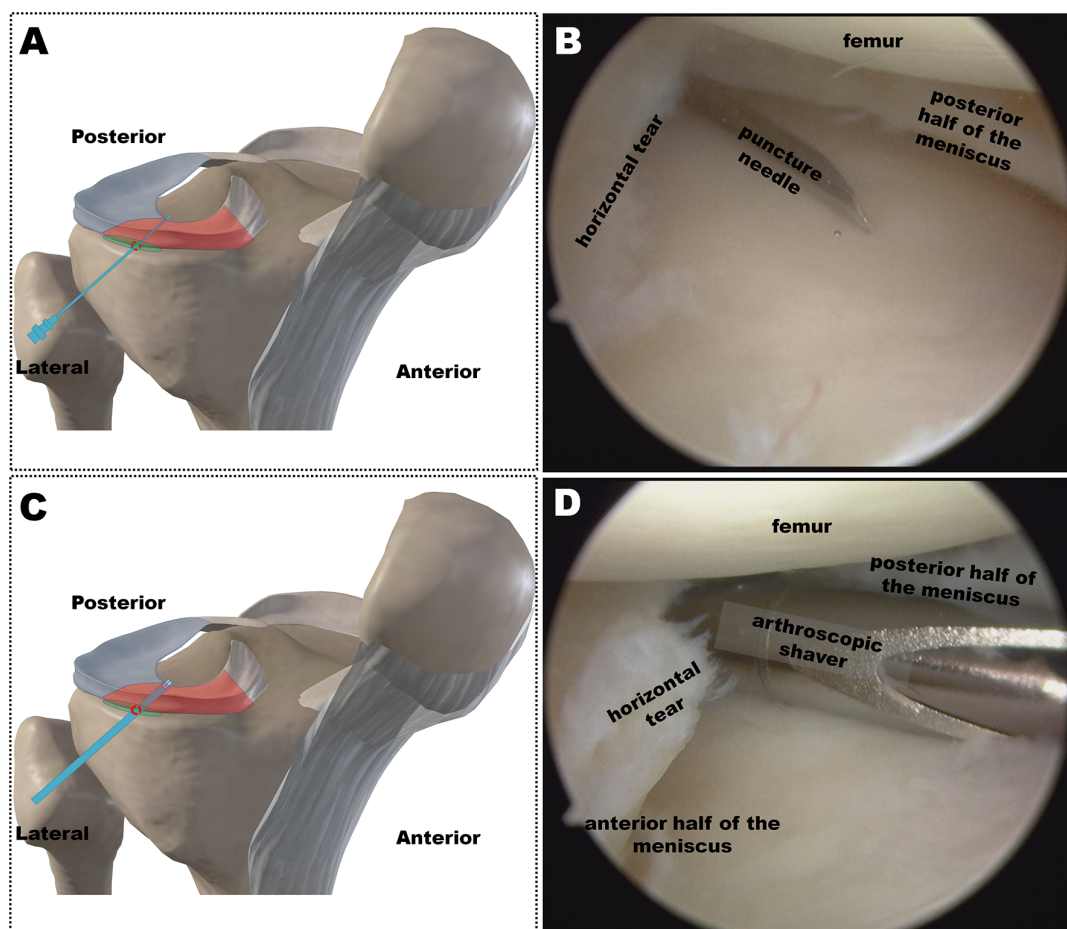


FIGURE 3 Establishment of the direct lateral inferior portal and subsequent debridement for a horizontal tear of the lateral meniscus in a right knee. (A) A long puncture needle is used for percutaneous localization, entering the space directly inferior to the lesion. (B) Arthroscopic view through the anterolateral portal showing the puncture needle successfully positioned between the inferior surface of the lateral meniscus and the tibial plateau. The needle is kept strictly parallel to the plateau to avoid iatrogenic damage to the meniscal parenchyma or articular cartilage. (C) After making a superficial skin incision and performing blunt dissection to protect the lateral stabilizing structures, an arthroscopic shaver is introduced through the newly established portal. (D) Arthroscopic monitoring from the anterolateral portal shows the shaver positioned directly inferior to the lateral meniscus, safely and effectively debriding the deep horizontal tear.

instruments, which often need to be prepared, such as 90° or 180° rongeurs. When performed concomitantly with more complex arthroscopic procedures like meniscal repair, this significantly prolongs operative time and compromises efficiency, posing a particular challenge for resource-limited operating rooms.

To overcome these challenges, the direct medial and lateral inferior portals described herein provide an optimal trajectory that significantly reduces surgical difficulty and eliminates the reliance on specialized instruments. Furthermore, this portal facilitates comprehensive inspection of the meniscal inferior surface, anterior and posterior horns, and root, aiding in the evaluation of surgical outcomes and the exclusion of occult injuries. Beyond debridement, it offers excellent visualization and working space for meniscal capsular margin and posterior root

repairs⁷ as well as the clearance of knee tophi or villonodular synovitis. The procedure is highly safe: it uses a standard 5-mm incision, and its establishment under direct arthroscopic visualization minimizes iatrogenic damage. As detailed in Table 2, the technique is highly versatile for both medial and lateral compartments. By strictly employing blunt dissection and targeting specific anatomical landmarks, it safely bypasses major structures like the lateral collateral ligament and anterolateral ligament. Despite its advantages, this technique has inherent limitations. First, it is a site-specific portal primarily indicated for horizontal tears in the anterior meniscal body and anterior horn, offering limited benefits for other lesion types or locations. Second, although performed under arthroscopic monitoring, there remains a potential risk of iatrogenic injury to the meniscal parenchyma or

TABLE 1 Pearls and Pitfalls of the Modified Inferior Horizontal Portal

Pearls
Use the area directly inferior to the meniscal tear or the junction of the anterior horn and body as the puncture reference point, and keep instruments parallel to the tibial plateau and inferior to the meniscus during the procedure to ensure accurate portal placement
After portal establishment, it can serve as either an operative or viewing channel and be combined with routine anterolateral and anteromedial portals to achieve comprehensive exploration and evaluation of the meniscal inferior surface, anterior/posterior horns, and roots
Perform both portal creation and tissue debridement under direct arthroscopic visualization to improve surgical efficiency and minimize iatrogenic injuries to the greatest extent
For the lateral portal, select the midlateral edge of the tibial plateau as the entry point. Use a scalpel only for a vertical skin incision, followed by blunt dissection to protect the LCL, ALL, and MCL
Pitfalls
Insufficient joint space opening during surgery is likely to increase the difficulty of instrument manipulation and the risk of cartilage damage
Using sharp dissection deep into the joint capsule or placing the lateral portal too posteriorly may increase the risk of iatrogenic injury to the LCL or ALL
ALL, anterolateral ligament; LCL, lateral collateral ligament; MCL, medial collateral ligament.

TABLE 2 Advantages and Disadvantages of the Modified Inferior Horizontal Portal

Advantages
It reduces surgical difficulty and the need for specialized instruments, and its minimally invasive incision does not increase the risk of postoperative complications, which is suitable for resource-limited settings
Full arthroscopic direct visualization during the entire procedure minimizes iatrogenic damage to the meniscus and articular cartilage, ensuring high surgical safety
It functions as a dual-purpose channel for both operation and observation, enabling comprehensive evaluation of the meniscus to exclude potential occult injuries
The technique is versatile for both medial and lateral menisci. By using blunt dissection and specific anatomical landmarks (e.g., midlateral edge for the lateral side), it safely avoids major ligamentous structures like the LCL and ALL
Disadvantages
It has narrow indications and is only applicable to deep horizontal tears in the anterior half of the meniscal body and anterior horn, with no value for other types or locations of meniscal injuries
Insufficient intraoperative joint space opening may increase instrument manipulation difficulty and the risk of damaging meniscal parenchyma or articular cartilage
The lateral compartment inherently has less laxity than the medial side, which may require more meticulous blunt dissection and careful instrument manipulation to avoid capsular or ligamentous strain
ALL, anterolateral ligament; LCL, lateral collateral ligament.

articular cartilage during portal establishment and instrument manipulation. To mitigate this, maximizing joint space opening via patient positioning is essential. Additionally, because the lateral compartment is inherently less lax than the medial side (Table 2), lateral procedures demand meticulous blunt dissection and careful manipulation to avoid capsular or ligamentous strain. Therefore, to balance safety and efficiency, we recommend this approach only when comprehensive arthroscopic

evaluation confirms that standard debridement is hindered and the lesion location is suitable.

DISCLOSURES

The authors (W.L., D.G., Z.L., X.L.) declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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